

SARDAR PATEL UNIVERSITY
M. Sc. (IT) INTEGRATED EXAMINATION, 1st SEM

Date : 16th NOV 2019, Saturday
 Session : Morning
 Course No : PS01CIIT31

Time : 10:00am To 01:00pm
 Sub: System Software
 Total marks : 70

Q – 1 Select an appropriate option

[08]

- i) _____ part of compilation breaks up the source program into essential pieces.
 a) Synthesis b) Analysis c) Execution d) None of these
- ii) The term _____ is used to represent the rules of meaning of a domain
 a) Execution b) Application c) Semantic d) Function
- iii) _____ data structure is used in synthesis phase.
 a) Symbol Table b) DFD c) Name Table d) All of above
- iv) Mnemonic has _____ primary fields.
 a) Mnemonic and opcode b) Name and Address
 c) Mnemonic and name d) Name and opcode
- v) The _____ of a program is the address of the instruction from which its execution must begin.
 a) Execution Address b) Address c) Execution start address d) All
- vi) _____ is the process of binding an external reference to the correct link time address.
 a) Translation b) Linking c) Loading d) None of above
- vii) _____ is responsible for interpreting user command and implementing them by invoking different modules of the application code.
 a) Dialog Manager b) Debug Manager c) Presentation manager d) All
- viii) _____ helps the user in selecting test data for his program.
 a) Program environment b) Language processor
 c) Software tools d) Test data generator

Q – 2 Answer the following Questions (Any SEVEN)

[14]

- i) What do you mean by Compiler
- ii) Draw a diagram of a language processing system.
- iii) Define system software and write types of software
- iv) What is Machine instructions format?
- v) What is Condition code? How it is tested?
- vi) Why origin of a program may have to be changed by the linker or loader?
- vii) Explain address.sensitive program
- viii) List down steps for testing and debugging.
- ix) Explain the factor on which efficiency of programs depends.

Q – 3 a) Explain software tools that manipulate source programs

[06]

b) List the phases of a compiler and explain symbol table management, error detection reporting.

[06]

OR

Q – 3 b) Give the schematic of interpretation of HLL program and execution of a machine language program by the CPU.

[06]

(1)

(P.T.O.)

- Q – 4 (a) What is assembly language? Explain the three basic facilities of assembly language. [06]
(b) What is assembly language? What kind of statements are present in an assembly language program? Discuss. Also highlight the advantages of assembly language. [06]
OR
- Q – 4 (b) Explain design specification of an assembler. [06]
- Q – 5 a) Explain Translated, Linked and Load time address. [06]
b) Explain different editors in detail. [06]
OR
- Q – 5 b) Discuss various steps for execution of program written in a programming language. [06]
- Q – 6 a) Which are the four basic categories in which programming environment can be classified? Explain each in brief. [06]
b) Briefly discuss the principles of command dialog design. [06]
OR
- Q – 6 b) Explain Debug Monitor and User Interfaces in detail [06]

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